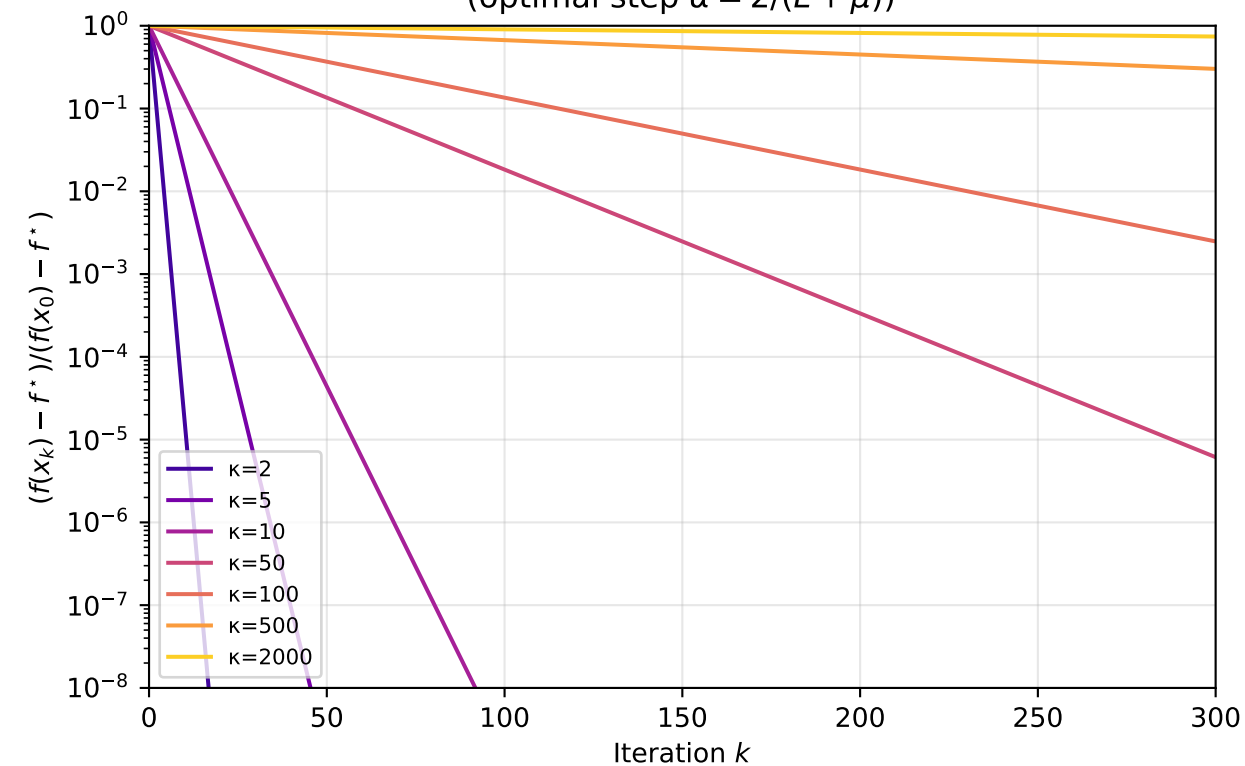
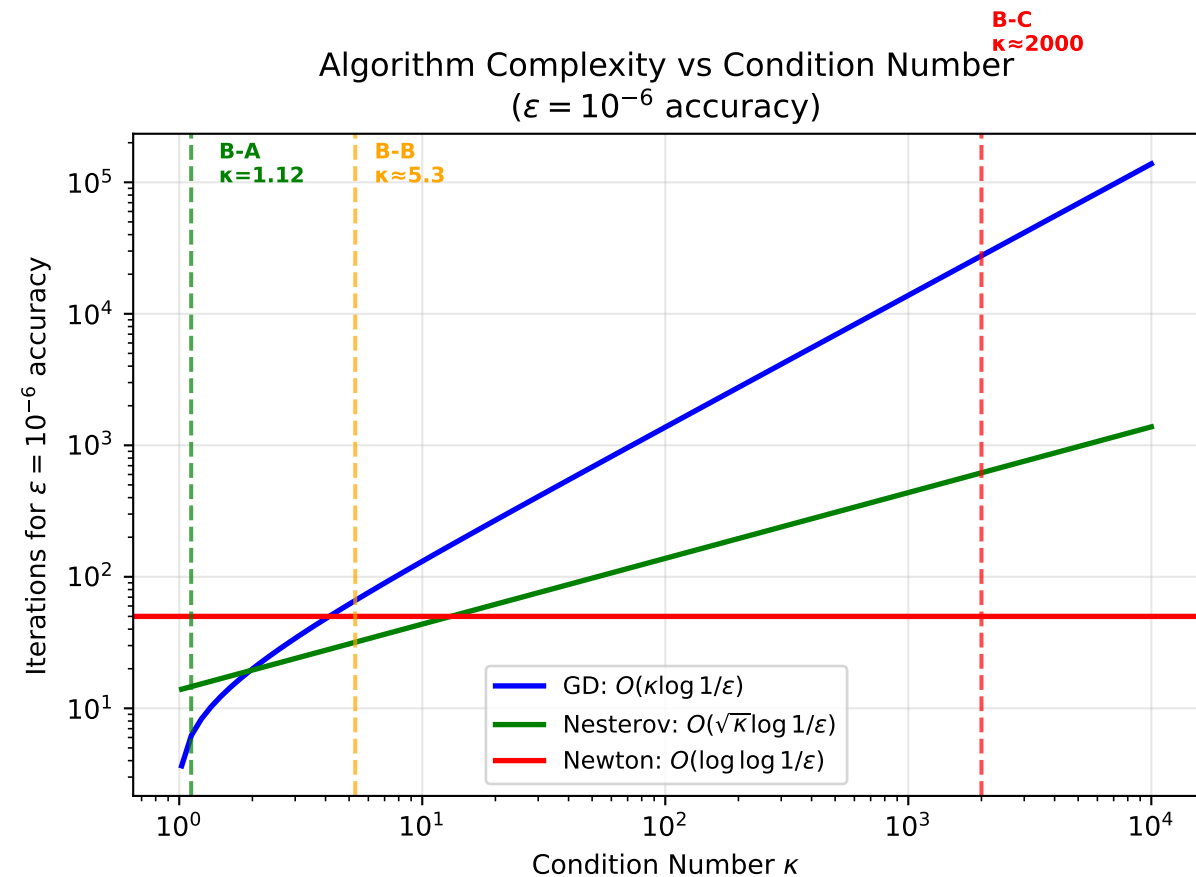


Theoretical Analysis: Condition Number as Central Determinant of Optimisation Algorithm Performance

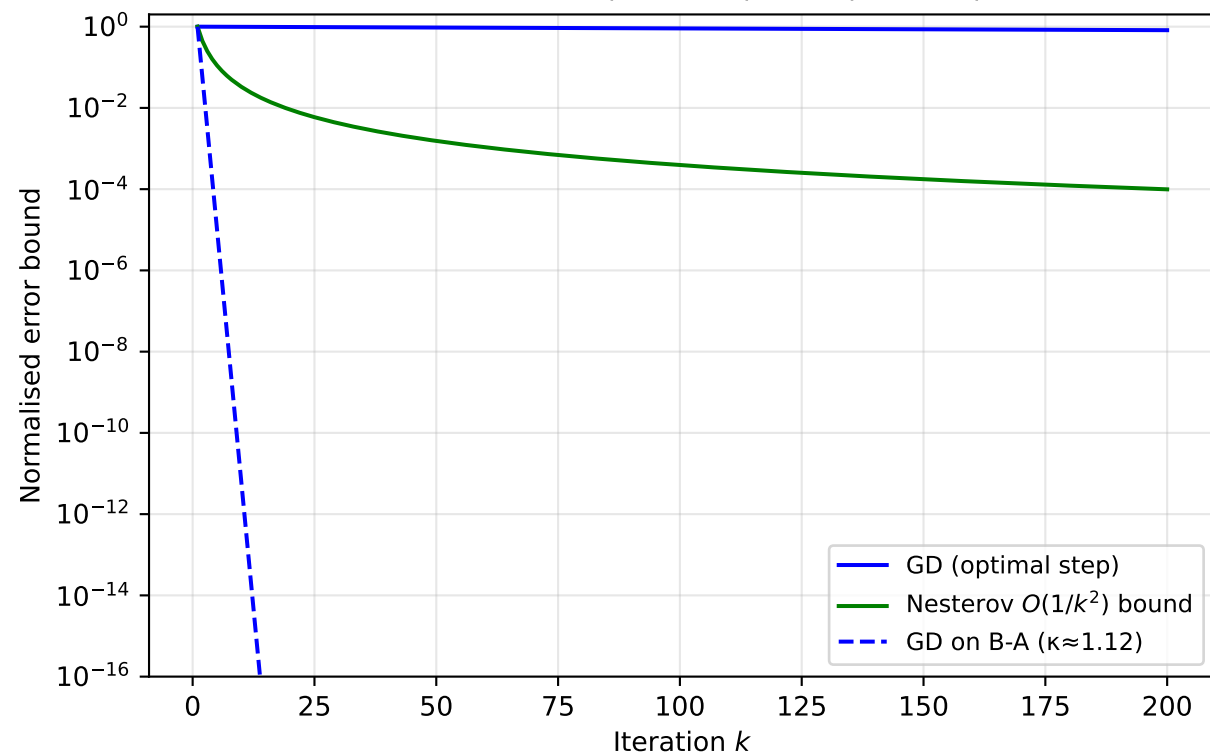
GD Linear Rate vs Condition Number (optimal step $\alpha = 2/(L + \mu)$)



Algorithm Complexity vs Condition Number ($\epsilon = 10^{-6}$ accuracy)



Convergence Rate Comparison Benchmark C ($\kappa=2000$) vs A ($\kappa \approx 1.12$)



Final Objective on Rosenbrock (All Methods, Various Iteration Counts)

